

PSTC LAB WEEK-01 REPORT

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Aim:

To learn the fundamental C programming by implementing programs using variables, operators, conditional statements, loops, and switch statement

Programs performed:

- 1) Even or odd This program checks whether a given number is even or odd using the modulus (%) operator and an if-else statement
- 2) Positive, Negative, or zero This program determines whether a number is +ve, -ve, or zero using relational operators and an if-else-if ladder.
- 3) Character type check This program identifies whether the entered character is an uppercase letter, lowercase letter, digit, or special character.
- 4) Leap Year check This program checks if a given year is a leap year by applying the standard leap year conditions with nested if statements.
- 5) Memory allocation of Data types: This program displays the memory size of different C data types using size of operator.
- 6) Menu-Based Arithmetic operations This program performs arithmetic operations such as addition, subtraction, multiplication, division, modulus, & Power using switch statement
- 7) Swap without temporary variable This program

Swaps two numbers using by storing one value using arithmetic operations without using an extra variable

8. Swap using temporary variable: This swaps two numbers by storing one value in temporary variable

9) Perfect square check: This program checks whether a number is a perfect square without using sqrt (library function)

10) Student Grade calculation: This program reads a student's marks and displays the corresponding grade using nested if-else statement or if else ladder

11) Extended menu - Based calculator: This program improves the calculator by handling invalid menu choices and division by zero.

12) Largest of three numbers: This program finds the largest among three numbers using only the conditional (ternary) operator without using if-else statement

13) Grade point to letter grade: This program is done by using switch statement and some relational checks.

Concepts Used:

Operators Used: 1) Arithmetic operators 2) Logical operator 3) Ternary operator

Control statements: 1) If 2) nested if 3) switch-case

Looping statements: 1) For loop 2) while loop

Conclusion:

All the programs were implemented, compiled, executed successfully, and the expected outputs were obtained.

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